FLORAPULSE SYSTEMS — PROJECT REPORT

Production Design: Iris Species Classification & FastAPI Gateway Integration

Project Title:	FloraPulse API Core	**Student Name:**	Anushay Ayat
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Current Status:	**Verified & Complete**	**Department:**	AI & Data Science

1. Executive Summary

Yeh technical report FloraPulse ecosystem ke automated real-time classification pipeline ko define karti hai. Is project ka main objective continuous structural dimensions (Sepal aur Petal metrics) ka use karke biological classes ko accurately differentiate karna aur use ek highly optimized REST API end-point ke zariye microservice architecture par deploy karna hai.

Infrastructure Verification: Pure backend workflow mein data constraints aur input parameter filtering ko secure karne ke liye dynamic Pydantic structures ka use kiya gaya hai.

2. Dataset Matrix & Dimensional Space

Is systems framework mein 4 primary continuous engineering inputs ko manage kiya gaya hai jo targeted flower taxonomy ko map karte hain:

Feature Vector	Data Parameter Type	Target Feature Significance
Sepal Length (cm)	Continuous Float64	Outer leaf longitudinal baseline metric layout.
Sepal Width (cm)	Continuous Float64	Lateral width scale mapping across outer structures.
Petal Length (cm)	Continuous Float64	High variance split index for immediate Setosa isolation.
Petal Width (cm)	Continuous Float64	Transverse distribution optimization index.

3. Predictive Modeling & Optimization

Classification algorithm ke liye ek robust Random Forest Ensemble model (100 independent estimators) use kiya gaya hai. Model generalizability score ko increase karne aur multi-class features space matrix ko secure karne ke liye data layers ko stratified 80/20 training aur testing matrix limits mein split kiya gaya hai.

4. Asynchronous FastAPI Gateway Service

Model configuration ko successfully train karne ke baad active state tensors ko Joblib implementation ke through serialize karke `.pkl` extension format mein save kiya gaya hai. FastAPI integration real-time HTTP protocols handle karne ke liye background runtime environment setup karti hai.

- **GET `/` :** Routing stability loop aur global status health check run karta hai.
- **POST `/predict` :** Multi-dimensional structural parameters ko parse karke live category mapping respond karta hai.

5. Request Payload Verification

FastAPI application test execution runs ke dauran user verification handshake ke liye standard target input framework check:

```
{
  "sepal_length": 5.1,
  "sepal_width": 3.5,
  "petal_length": 1.4,
  "petal_width": 0.2
}
```